

Q I have a 1.5 GHz CPU with an Intel 845GBV motherboard, a 40 GB hard disk, an nVidia GeForce4 MX 440 graphics card and a Creative Sound Blaster Live DE 5.1 sound card. Can I use another fan for the graphics card instead of its own fan? If so, what kind of fan should I use? Will it enhance my card's performance?

Udayan

A The fan that comes with the card is more than enough! You need not change or add any more fans to it. However, you can increase the number of fans in the cabinet, as it might improve the cooling system, and thereby increase the lifespan of the component. However, make sure you have a better fan to keep things cool in case you wish to over clock your graphics card for better performance.

Q I have three questions—what is a firmware update, and what does it do? Which of the following needs regular driver and firmware updates for optimal functionality: motherboard, chipset, graphics card, sound card, hard disk, processor, modem, monitor, or the TV tuner card? Lastly, I intend buying a PC, primarily to play games. Which OS should I opt for—Windows XP Professional or Windows XP Home Edition?

Akhil Bahri

A Firmware is software that's embedded in the hardware. It's accessible through specially written applications that can write or read from the ROM chip, where the firmware resides. For instance, the BIOS on the motherboard is the firmware. As and when manufacturers discover bugs in the hardware, they put up specific updates to overcome them. Firmware updates can also help overcoming hardware limitations.

Assume that your current AMD board supports a 166 FSB processor. Later, if AMD releases a 200 FSB CPU, then the motherboard manufacturer may release a firmware update so that your board also supports it. In the case of such cards as TV-tuners, the manufacturer may release an update to support a new mother-